Dream11 — Next-Gen Team Builder with Predictive AI

Feature Documentation

Inter IIT Tech Meet 13.0 Dataset: `final_featured_v2` (cleaned)

Overview

Property	*Value*
Total rows (raw)	*483,612*
Total rows (cleaned)	*480,891*
Rows dropped	*2,721 (0.56%) — missing `team` / `opposition`*
Total features	*63*
Target variable	`fantasy_points`
Training cutoff	*2024-06-30 (strict — no data after this date used in training)*
Data source	*Cricsheet*

*Each row represents **one player's performance in one match**. All rolling/historical features are computed using only past data (shifted by 1) to prevent data leakage.*

Feature Groups

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1. Identity & Match Context

*These columns are **not used as model input features** — they serve as identifiers and metadata for joining, filtering, and evaluation.*

Feature	Type	Description
`match_id`	*int*	*Unique Cricsheet match identifier. Used for grouping players per match during team selection.*
`date`	*datetime*	*Date of the match. Used to enforce training cutoff (≤ 2024-06-30) and for rolling window computation.*
`venue`	*string*	*Name of the stadium/ground. Used to compute venue-level aggregated features.*
`match_type`	*string*	*Format of the match — e.g.* `T20`, `ODI`, `Test`.
`team`	*string*	*Team the player represents in this match. Must have at least 1 player per team in the final Dream11 XI.* ***Rows with null*** **`team`** ***have been dropped.***
`opposition`	*string*	*The opposing team in this match.* ***Rows with null*** **`opposition`** ***have been dropped.***
`player`	*string*	*Player name as listed in Cricsheet data. Used as the display identifier in Product UI output.*
`role`	*string*	*Player's primary role — one of* `BAT`, `BOWL`, `AR` *(All-Rounder),* `WK` *(Wicket-Keeper). Used to enforce Dream11 team composition constraints during selection.*
`format_encoded`	*int*	*Numerical encoding of* `match_type`*. Allows the model to treat format as an ordinal/categorical signal. 0 is used for ODI, 1 for ODM and test/MDM have 2.*

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2. Target Variable

Feature	***Type***	***Description***
`fantasy_points`	*float*	*Actual fantasy points scored by the player in this match, computed per Dream11's official point system. This is what the model is trained to predict.*

Fantasy Point System (summary):

- *Batting: runs, boundaries, strike rate bonuses, fifties, hundreds*
- *Bowling: wickets, maidens, economy rate bonuses, 3-wicket/5-wicket hauls*
- *Fielding: catches, stumpings, run-outs*
- *Dismissal penalty: duck (batsmen)*

> ***Note:*** *At prediction time (Product UI), `fantasy_points` is unknown — the model predicts it from historical and contextual features only.*

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3. Batting Features

*Rolling averages are computed over the player's last N **innings** (not matches), shifted by 1 to exclude the current match, preventing leakage.*

Feature	***Type***	***Description***
`runs_avg_3`	*float*	*Average runs scored over last 3 innings. Short-term batting form indicator.*
`runs_avg_5`	*float*	*Average runs scored over last 5 innings. Medium-term batting form.*
`runs_avg_10`	*float*	*Average runs scored over last 10 innings. Long-term batting consistency.*
`sr_avg_3`	*float*	*Average strike rate over last 3 innings. Captures recent scoring tempo.*
`sr_avg_5`	*float*	*Average strike rate over last 5 innings.*
`sr_avg_10`	*float*	*Average strike rate over last 10 innings.*
`boundary_runs`	*int*	*Total runs scored via boundaries (4s and 6s) in this match. Proxy for aggressive batting.*

boundary_percent	*float*	*Proportion of runs scored via boundaries. Indicates batting style.*
runs_std_5	*float*	*Standard deviation of runs over last 5 innings. Measures batting consistency — low value means reliable, high value means volatile.*
runs_std_10	*float*	*Standard deviation of runs over last 10 innings.*
fifty_plus	*int*	*Binary flag — 1 if the player scored 50+ runs in the current match, else 0.*
hundred	*int*	*Binary flag — 1 if the player scored 100+ runs in the current match, else 0.*
fifty_rate_5	*float*	*Rate of scoring 50+ in last 5 innings (0 to 1). Captures big-innings frequency.*
fifty_rate_10	*float*	*Rate of scoring 50+ in last 10 innings.*
balls_per_boundary	*float*	*Average balls faced per boundary hit. Lower = more aggressive.*
dots	*int*	*Number of dot balls faced in this match. Supplementary batting activity signal.*
is_out	*int*	*Binary — 1 if player was dismissed, 0 if not out.*
dismissal_kind	*string*	*Mode of dismissal (caught, bowled, run out, etc.). Not used as direct model input — informational.*
extras	*int*	*Extras conceded by bowling team. Contextual field, primarily relevant for bowling analysis.*

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4. Bowling Features

Feature	*Type*	*Description*
overs	*float*	*Overs bowled by the player in this match.*
bowling_sr	*float*	*Bowling strike rate — balls bowled per wicket.*

wickets_avg_3	float	*Average wickets taken over last 3 bowling appearances.*
wickets_avg_5	float	*Average wickets taken over last 5 bowling appearances.*
wickets_avg_10	float	*Average wickets taken over last 10 bowling appearances.*
wickets_std_5	float	*Standard deviation of wickets over last 5 appearances. Captures bowling consistency.*
wickets_std_10	float	*Standard deviation of wickets over last 10 appearances.*
economy_avg_3	float	*Average economy rate over last 3 bowling appearances. Lower = more economical.*
economy_avg_5	float	*Average economy rate over last 5 bowling appearances.*
economy_avg_10	float	*Average economy rate over last 10 bowling appearances.*
economy_std_5	float	*Standard deviation of economy rate over last 5 appearances.*
three_wicket_haul	int	*Binary — 1 if player took 3+ wickets in this match.*
five_wicket_haul	int	*Binary — 1 if player took 5+ wickets in this match.*
maiden_overs	int	*Number of maiden overs bowled in this match.*
dot_ball_percent	float	*Proportion of balls bowled that were dot balls. Higher = more economical/threatening bowler.*

> ***Note:*** *Rolling bowling features default to 0 for players with no recent bowling history. This is expected and correct — batsmen with no bowling record will have zeros here.*

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5. Fielding Features

Feature	Type	Description
fielding_points_raw	int	*Raw fielding points earned in this match (catches, stumpings, run-outs).*

`fielding_avg_3`	*float*	*Average fielding points over last 3 matches.*
`fielding_avg_5`	*float*	*Average fielding points over last 5 matches.*
`fielding_avg_10`	*float*	*Average fielding points over last 10 matches.*

> *Fielding is often underweighted in fantasy models but can be decisive for WK and close-in fielders.* `fielding_avg_5` *tends to be the most predictive window.*

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6. Fantasy Form Features

These features operate directly on the fantasy points time series rather than raw cricket statistics. They capture overall form holistically.

Feature	Type	Description
`fantasy_avg_3`	*float*	*Average fantasy points over last 3 matches. Best short-term form signal.*
`fantasy_avg_5`	*float*	*Average fantasy points over last 5 matches.*
`fantasy_avg_10`	*float*	*Average fantasy points over last 10 matches. Long-term form baseline.*
`fantasy_std_5`	*float*	*Standard deviation of fantasy points over last 5 matches. Measures how consistent the player's overall fantasy returns are.*
`fantasy_std_10`	*float*	*Standard deviation of fantasy points over last 10 matches.*
`form_momentum`	*float*	*Difference between short-term and long-term fantasy average (e.g.* `fantasy_avg_3` − `fantasy_avg_10`*). Positive = player in rising form, negative = declining form.*

> `fantasy_avg_3` *is typically the single strongest predictor of next-match fantasy points.* `form_momentum` *adds directional signal on top of raw averages.*

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7. Career & Contextual Features

Feature	Type	Description
`career_matches`	*int*	*Total number of matches played by the player in the dataset. Proxy for experience. Low values indicate sparse history — predictions less reliable.*
`career_avg_fp`	*float*	*Career average fantasy points across all historical matches. Baseline expectation for the player.*
`days_since_last_match`	*int*	*Number of days since the player last appeared in a match. Captures availability and match rhythm. High values may indicate injury/rest.*

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8. Venue & Opposition Features

These contextual features capture how external conditions influence fantasy point returns.

Feature	Type	Description
`venue_avg_fp`	*float*	*Historical average fantasy points at this venue across all players.*
`venue_format_avg_fp`	*float*	*Average fantasy points at this venue broken down by match format (T20/ODI/Test). More granular and reliable than `venue_avg_fp`.*
`venue_high_scoring`	*int*	*Binary 1 if this venue is historically high-scoring, 0 otherwise.*
`vs_team_avg_fp`	*float*	*Player's historical average fantasy points when playing against this specific opposition. Captures head-to-head track record.*
`opp_batting_strength`	*float*	*Average rolling batting performance (bat_avg_5) of the opposition team's batsmen. Higher = stronger batting lineup the player is bowling/fielding against.*
`opp_bowling_strength`	*float*	*Average rolling bowling performance (bowl_avg_5) of the opposition team's bowlers. Higher = tougher bowling attack the player is batting against.*

`opp_batting_strength` and `opp_bowling_strength` are computed using the opposition team's players' rolling averages at the time of the match — no leakage. Null values are filled with the dataset mean.

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9. Data Quality Notes

Dropped Rows

- ***2,721 rows dropped** where both `team` and `opposition` were null*
- *These were always co-null (same rows) — confirmed*
- *Represents **0.56% of total data** — negligible impact on model training*
- *Cleaned dataset: **480,891 rows***

Possibly High Correlation Groups (for feature selection)

The following groups may be internally highly correlated. During model training, pay attention to these groups:

- ***Runs rolling:** `runs_avg_3`, `runs_avg_5`, `runs_avg_10`*
- ***Wickets rolling:** `wickets_avg_3`, `wickets_avg_5`, `wickets_avg_10`*
- ***Economy rolling:** `economy_avg_3`, `economy_avg_5`, `economy_avg_10`*
- ***Fantasy rolling:** `fantasy_avg_3`, `fantasy_avg_5`, `fantasy_avg_10`*

Training Cutoff Compliance

*Per the problem statement, **no data after 2024-06-30 may be used for training**. All rolling features are computed with a lag of 1 (shifted), so the feature for a match on date D uses only data from matches before D. The training cutoff must be enforced at the `date` column level when splitting data.*

Feature Summary Table

#	Feature	Group	Type
1	match_id	Identity	int
2	date	Identity	datetime
3	venue	Identity	string

4	*match_type*	*Identity*	*string*
5	*team*	*Identity*	*string*
6	*opposition*	*Identity*	*string*
7	*player*	*Identity*	*string*
8	*role*	*Identity*	*string*
9	*format_encoded*	*Identity*	*int*
10	*fantasy_points*	***Target***	*float*
11	*dots*	*Batting*	*int*
12	*dismissal_kind*	*Batting*	*string*
13	*is_out*	*Batting*	*int*
14	*extras*	*Batting*	*int*
15	*runs_avg_3*	*Batting*	*float*
16	*runs_avg_5*	*Batting*	*float*
17	*runs_avg_10*	*Batting*	*float*
18	*sr_avg_3*	*Batting*	*float*
19	*sr_avg_5*	*Batting*	*float*
20	*sr_avg_10*	*Batting*	*float*
21	*boundary_runs*	*Batting*	*int*
22	*boundary_percent*	*Batting*	*float*
23	*runs_std_5*	*Batting*	*float*
24	*runs_std_10*	*Batting*	*float*
25	*fifty_plus*	*Batting*	*int*
26	*hundred*	*Batting*	*int*
27	*fifty_rate_5*	*Batting*	*float*
28	*fifty_rate_10*	*Batting*	*float*

29	*balls_per_boundary*	*Batting*	*float*
30	*overs*	*Bowling*	*float*
31	*bowling_sr*	*Bowling*	*float*
32	*wickets_avg_3*	*Bowling*	*float*
33	*wickets_avg_5*	*Bowling*	*float*
34	*wickets_avg_10*	*Bowling*	*float*
35	*wickets_std_5*	*Bowling*	*float*
36	*wickets_std_10*	*Bowling*	*float*
37	*economy_avg_3*	*Bowling*	*float*
38	*economy_avg_5*	*Bowling*	*float*
39	*economy_avg_10*	*Bowling*	*float*
40	*economy_std_5*	*Bowling*	*float*
41	*three_wicket_haul*	*Bowling*	*int*
42	*five_wicket_haul*	*Bowling*	*int*
43	*maiden_overs*	*Bowling*	*int*
44	*dot_ball_percent*	*Bowling*	*float*
45	*fielding_points_raw*	*Fielding*	*int*
46	*fielding_avg_3*	*Fielding*	*float*
47	*fielding_avg_5*	*Fielding*	*float*
48	*fielding_avg_10*	*Fielding*	*float*
49	*fantasy_avg_3*	*Fantasy Form*	*float*
50	*fantasy_avg_5*	*Fantasy Form*	*float*
51	*fantasy_avg_10*	*Fantasy Form*	*float*
52	*fantasy_std_5*	*Fantasy Form*	*float*
53	*fantasy_std_10*	*Fantasy Form*	*float*

54	*form_momentum*	*Fantasy Form*	*float*
55	*career_matches*	*Career*	*int*
56	*career_avg_fp*	*Career*	*float*
57	*days_since_last_match*	*Career*	*int*
58	*venue_avg_fp*	*Venue*	*float*
59	*venue_format_avg_fp*	*Venue*	*float*
60	*venue_high_scoring*	*Venue*	*int*
61	*vs_team_avg_fp*	*Opposition*	*float*
62	*opp_batting_strength*	*Opposition*	*float*
63	*opp_bowling_strength*	*Opposition*	*float*

Prepared for Inter IIT Tech Meet 13.0 — Dream11 Problem Statement Dataset validated and cleaned by documentation team Cricsheet data source: https://cricsheet.org/